

Dissertation on

"Coral Reef Bleaching Risk Prediction Using Satellite Alerts and NLP on Observation Reports"

Submitted in partial fulfillment of the requirements for

Bachelor of Technology
in
Computer Science & Engineering

UE23CS320A – Capstone Project Phase - 1

Submitted by:

Sakshi M Pai	**PES2UG23CS514**
Sampada Joshi	**PES2UG23CS521**
Anushka Mandal	**PES2UG23CS083**
Divya J	**PES2UG24CS810**

Under the guidance of

Dr. M Farida Begam
Professor
PES University

Aug-Dec 2025

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FACULTY OF ENGINEERING
PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

FACULTY OF ENGINEERING

CERTIFICATE

This is to certify that the dissertation entitled

‘Coral Reef Bleaching Risk Prediction Using Satellite Alerts and NLP on Observational Reports’

is a bonafide work carried out by

Sampada Joshi	**PES2UG23CS521**
Sakshi M Pai	**PES2UG23CS514**
Anushka Mandal	**PES2UG23CS083**
Divya J	**PES2UG24CS810**

In partial fulfillment for the completion of Fifth-semester Capstone Project Phase - 1 (UE23CS320A) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Aug 2025 – Dec. 2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 5th-semester academic requirements in respect of project work.

Signature	Signature
Dr. M Farida Begam	Dr. Sandesh B J
Professor	Chairperson

External Viva

Name of the Examiners	**Signature with Date**
1.Dr. M Farida Begam	_____
2.Ms. Umme Haani	
3.Dr. R Nandhi Kesavan	_____

DECLARATION

We hereby declare that the Capstone Project Phase - 1 entitled **"Coral Reef Bleaching Risk Prediction Using Satellite Alerts and NLP on Observation Reports"** has been carried out by us under the guidance of **Dr. M Farida Begam, Professor** and submitted in partial fulfillment of the course requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Aug. – Dec 2025. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

PES2UG23CS521	**SAMPADA JOSHI**
PES2UG23CS514	**SAKSHI M PAI**
PES2UG23CS083	**ANUSHKA MANDAL**
PES2UG24CS810	**DIVYA J**

ACKNOWLEDGEMENT

I would like to express my gratitude to **Dr. M Farida Begam,** Department of Computer Science and Engineering, PES University, for her continuous guidance, assistance, and encouragement throughout the development of this UE23CS320A - Capstone Project Phase – 1.

I am grateful to all Capstone Project Coordinators, for organizing, managing, and helping with the entire process.

I take this opportunity to thank Dr. Sandesh B J, Professor & Chairperson, Department of Computer Science and Engineering, PES University, for all the knowledge and support I have received from the department. I would like to thank Dr. B.K. Keshavan, Dean of Faculty, PES University for his help.

I am deeply grateful to Late Dr. M. R. Doreswamy, Founder, PES University, whose vision and dedication continue to inspire generations of learners. I would also like to express my sincere gratitude to Prof. Jawahar Doreswamy, Chancellor, PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University and Prof. Nagarjuna Sadineni, Pro Vice-Chancellor, PES University for providing to me various opportunities and enlightenment every step of the way. Finally, this project could not have been completed without the continual support and encouragement I have received from my family and friends.

ABSTRACT

Coral reef bleaching is a major environmental threat driven by rising sea surface temperatures and changing ocean conditions, yet traditional monitoring method such as satellite observations and manual field reports, often fail to provide early and accurate warnings. This project aims to develop a multimodal machine learning model that integrates environmental image data (SST, DHW, TSA) with textual insights extracted through Natural Language Processing from reef monitoring reports that describe early bleaching signs, color loss, coral cover changes, water quality, disease presence, and other stress indicators. By combining quantitative and qualitative information, the system enhances the reliability and timeliness of bleaching risk prediction, enabling conservation teams to respond proactively.

The study also investigates how these two types of data interact and which environmental or field-based features contribute most to the model's predictive power. The research workflow includes satellite data cleaning, NLP-based text preprocessing, feature extraction, multimodal fusion, and detailed model evaluation across different reef zones. A key objective is to design a framework that is not only accurate but also practical and adaptable, so that reef monitoring teams can integrate it into their existing workflows. By providing earlier and more context-aware alerts, the system aims to support long-term conservation planning and improve the overall resilience of coral reef ecosystems.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1.	INTRODUCTION	01
2.	PROBLEM STATEMENT	03
3.	LITERATURE SURVEY	05
4.	RESEARCH GAP	07
5.	OBJECTIVES	08
6.	Overview of Datasets	09
7.	CONCLUSION OF CAPSTONE PROJECT PHASE - 1	11
8.	PLAN OF WORK FOR CAPSTONE PROJECT PHASE - 2	12
REFERENCES/BIBLIOGRAPHY		13
APPENDIX A DEFINITIONS, ACRONYMS, AND ABBREVIATIONS		14

CHAPTER 1

INTRODUCTION

1.1 Coral Reefs and Bleaching

Coral reefs are one of the most diverse ecosystems on the planet, which provide critical habitat for marine ecosystems. They are often referred to as “rainforests of the sea”. Although they cover less than 1% of the ocean, yet they support approximately 25% of the marine species, because they provide essential habitats for fishes, and other marine organisms. Coral reefs support a variety of types of marine organisms including various types of fishes, invertebrates, crustaceans and other marine organisms. Many of these organisms are mainly dependent on coral reefs for their survival, shelter, breeding grounds and feeding. The complexity of these reefs creates microhabitats that can hold a large number of species, making them the hotspots of marine biodiversity.

Beyond this, they are also important for human society. They are a major source of food through fisheries, hence helping millions of people who rely only on reef based fisheries and also depend on invertebrates for primary protein source. Additionally, coral reefs also help improve tourism by providing a major source of recreational activities, which also generate millions of dollars annually. Divers, nature enthusiasts and also local stakeholders are attracted to the pretty and beautiful, colorful reefs. Coral reefs also play a significant role in predicting the arrival of various natural disasters, like tsunami, flood, deforestation etc. By buffering the impact of waves, storms, and tsunamis, reefs help reduce coastal erosion and property damage, safeguarding vulnerable coastal populations and infrastructure. Their ecological and economic value is thus irreplaceable, making their conservation a global priority.

Despite their significant importance, coral reefs are under huge threat because of climate changes, pollution, increasing sea surface temperature, and other thermal stressors. One of the most significant and visible impacts of these environmental stresses is coral bleaching.

Coral bleaching occurs because of the stress caused by the rising temperatures of the sea, pollution which expel the inner symbiotic algae (zooxanthellae) living in their tissues. This algae provides corals with their colors, through photosynthesis where the energy is gained. The loss of zooxanthellae causes the corals to turn white, hence called bleaching and weakens the coral, leaving them sensitive to diseases, which does not allow them to grow and nourish properly. Bleaching events are currently going on in many parts of the world's oceans, with the major severity recorded in 1998, 2010, 2015-2016, and 2020, which highlights the importance of early prediction of the bleaching.

1.2 Importance of prediction over detection

Traditional coral reef monitoring systems often rely on detection rather than prediction, usually through post-event satellite imagery. While detection is equally important for documenting the severity and extent of bleaching events, it is too late to help prevent this bleaching. By the time the bleaching has occurred, significant damage may have already taken place, which limits any conservation efforts and remediation measures.

Therefore, rather than detection or determining, prediction of coral bleaching events before they take place provides a greater edge for reef conservation. Predictive models can identify areas of high risk reef bleaching based on the various environmental factors such as sea surface temperature anomalies,

Chlorophyll, Degree Heating Weeks (DHW), Hotspots, Sea Salinity, and various other oceanographic parameters. Early warning of bleaching events helps conservationists and local stakeholders to take various measures, which further helps prevent bleaching actions in the future.

Prediction also enables integration of historical data and in-site observations, which improves accuracy and reliability of the prediction systems. Early detection of bleaching events help conservationists and local stakeholders to take effective steps to prevent further bleaching events. These measures may include restricting fishing measures, water quality management and targeted restoration efforts. Over time, predictive approaches can

also be used to model the long-term impacts of climate change on reef ecosystems, helping scientists and managers develop adaptive strategies for sustainable reef management.

CHAPTER 2

PROBLEM STATEMENT

Coral Reef Bleaching is currently one of the most severe environmental issues which causes severe threats to marine ecosystems, particularly in sensitive areas like Great Barrier reef, where the richness of available coral is noted. Rising sea temperatures, changing salinity and poor water quality have increased the stress on coral reefs, which lead to large scale reef bleaching. Existing and traditional monitoring and prediction systems, such as those using the parameters like Degree Heating Weeks (DHW), Sea Surface Temperature (SST), focus primarily on thermal stress indicators. These methods are effective to some extent. But these systems fail to capture the other significant biological, chemical and human-induced stress factors that influence reef health. They neglect the real time qualitative data contained in the observational reports, given by various marine institutes. These data contain the actual reef surveys and sea conditions, which describe the early signs of bleaching or changes in coral conditions, which are equally effective for predicting coral bleaching.

The existing prediction systems show either low accuracy or do not use a proper methodology, due to the lack of integration between qualitative satellite alerts and unstructured observational reports.

Additionally, current approaches mostly only identify bleaching after a significant coral bleaching event has occurred, rather than proactively predicting the bleaching risk. Therefore, there is an urgent need to create a multimodal prediction framework that can combine real-time satellite alerts with textual information from observational field reports to provide more accurate and comprehensive and context-aware information of coral reef bleaching risk.

This project aims to bridge the gap by creating a machine learning model that integrates environmental parameters from satellite alerts along with Natural Language Processing (NLP)-based features extracted from observational reports to predict coral bleaching risk more accurately for the Great Barrier Reef region.

CHAPTER 3

LITERATURE SURVEY

Understanding coral bleaching has become increasingly important because reefs around the world are facing more frequent temperature stress and environmental change. Recent research has explored satellite data, drones, underwater imaging, and machine learning to better detect and predict bleaching. The following survey summarizes seven studies published between 2024 and 2025 that focus on coral reef bleaching detection, monitoring, and environmental drivers.

The study by **Yang, Liu, and Xu (2024)** focuses on using multitemporal Sentinel-2 satellite images to detect bleaching. The authors highlight that bleaching signals are usually very weak in optical satellite imagery because the water column and atmospheric effects distort the signal. To overcome this, they compare images from different dates and apply corrections before classifying bleaching areas. Their work shows that Sentinel-2 can capture bleaching patterns, but only when good preprocessing and multitemporal comparison are used due to the subtle nature of bleaching reflectance.

In another approach, **Besson, Nebiker, and Kunz (2024)** use UAV imagery to map coral reefs through object-based image analysis. Instead of pixel-based methods, they segment the images into objects and classify them. Because drones can fly close to the water and capture high-resolution imagery, their method detects fine-scale changes that satellites usually miss. This study shows that drone-based OBIA is promising for local-level reef monitoring, especially when high detail is required.

Similarly focusing on imagery, **Alzahrani, Almutairi, Alotaibi, and Alshamrani (2024)** explore deep learning techniques for classifying coral health using enhanced underwater

images. Their approach improves raw underwater photographs using image enhancement techniques before feeding them to CNN models. The improved visibility helps the model classify coral conditions more accurately. Their findings suggest that enhanced imagery significantly boosts the performance of deep learning models, making it useful for diver-based or robot-based surveys.

Moving into predictive modelling, **Shaha, Rathod, More, and Shelke (2024)** investigate how environmental factors can be used to predict bleaching. Their machine learning model includes variables like sea surface temperature (SST), degree heating weeks (DHW), salinity, and sunlight exposure. They show that thermal stress indicators, especially SST and DHW, play the most dominant role in predicting bleaching events. Their work highlights that ML models can be powerful tools for early warnings if enough environmental data is available.

Focusing more on ecology, **González-Rivero, Hoegh-Guldberg, Bongaerts, Cortés, and Dove (2024)** study the environmental drivers behind bleaching when corals face thermal stress. Their research shows that not only temperature but also light levels, water clarity, and local reef conditions can influence how severely corals bleach. The authors emphasize that bleaching is not solely caused by heat; instead, it results from a combination of interacting stressors, which explains why some reefs bleach more than others even under similar temperatures.

On a longer timescale, **Hensel, Oliver, Burden, Grossman, Felis, and Grottoli (2025)** examine how long-term climate variability shapes bleaching patterns. They analyze historical climate trends and find that large-scale cycles, such as ENSO and long-term warming trends, strongly influence when and where bleaching happens. Their study highlights that understanding climate cycles is crucial for long-term coral reef management and for predicting future bleaching occurrences.

Lastly, **Ichikawa, Omori, Iwasaki, Yabe, and Sato (2024)** combine environmental indicators with multispectral analysis to assess bleaching. Their approach uses a mix of

satellite-derived parameters — such as chlorophyll, turbidity, SST, and multispectral reflectance — to evaluate coral conditions. By merging environmental datasets with spectral observations, they show that bleaching assessments become more reliable. This integrated method provides a more complete picture of reef health compared to relying on temperature alone.

CHAPTER 4

RESEARCH GAP

Current Coral bleaching predictions have several significant issues. Most of the existing data are over relying on satellite data, and most of the models are majoring on only one feature, which is sea surface temperature, while other factors remain unutilized. The other factors are equally important in playing a significant role in coral bleaching, such as chlorophyll, ocean color, pH, salinity of the ocean etc.

In terms of methodology and machine learning algorithms, there is a clear need for usage of advanced models. The coral reef bleaching has patterns in them. So, the basic models are not suitable for detecting these types of complex patterns.

The integration of observational reports using natural language processing is still far limited, and the structured and unstructured data are weekly combined, which significantly reduces the model's accuracy.

In addition to this, many studies have clearly shown signs of over reliance on only one factor called Sea Surface Temperature (SST). This leaves gaps in understanding the other important factors which are also responsible for the bleaching process. The other environmental factors like chlorophyll, pH, salinity as well as, the satellite alerts factors too like Degree Heating Weeks (DHW), HotSpot (HS), Bleaching Alert Area (BAA), remain unexplored.

There is a notable lack of real-time prediction or near real-time prediction system, which is most important, as this does not help the conservationists, and reef managers in order to take the future steps for reef protection. Collectively, these limitations highlight the need for more comprehensive, integrated, and timely predictive approaches for coral reef conservation.

CHAPTER 5

OBJECTIVES

The main aim of this project is to build a system that can predict the risk of coral reef bleaching by combining two kinds of information: satellite images that capture heat-stress indicators and written reports that contain on-site observations. Since the model uses more than one type of data, it becomes a multimodal approach rather than relying on a single source. One of the ideas behind this work is to understand how the descriptive reports written by marine institutes can actually strengthen the satellite-based alerts. These reports often mention details like unusual water temperature, changes in salinity, pH variations, or early color fading—things that satellite data usually cannot capture directly.

Along with developing the model, the project also tries to examine how dependable the current methods really are and where their limitations appear. Another important goal is to check whether the proposed model can consistently predict bleaching risk across different locations by evaluating it with suitable performance measures. The study also tries to make the prediction process more understandable by identifying the factors that have the strongest influence on bleaching. Finally, the work aims to present the outcomes in a clear and usable way so that reef conservation teams can make practical decisions based on the results. A longer-term objective is to help move toward more standardized and reproducible workflows in coral-reef monitoring, giving future researchers a stronger base to build upon.

CHAPTER 6

OVERVIEW OF DATASETS

This project uses a multimodal dataset consisting of two primary sources:
(1) satellite-derived heat-stress indicators
(2) textual observational reports.

Both datasets are aligned by date and region to support bleaching risk prediction.

6.1 Satellite Heat-Stress Images (NOAA Coral Reef Watch)

Satellite data were collected from the *NOAA Coral Reef Watch – Archived Data and Images* portal, which provides historical coral reef monitoring products. Using this interface, the Great Barrier Reef region was selected and daily maps were downloaded by specifying the required year, month and day, The platform offers several coral-relevant variables; in this study, the following are used:

9. Sea Surface Temperature (SST)
10. SST Anomaly (SSTA)
11. Coral Bleaching HotSpot (HS)
12. Degree Heating Week (DHW)
13. Bleaching Alert Area (BAA)

Each map is provided as a static GIF image containing spatial,color-coded heat-stress information for the selected date.These images serve as the environmental modality of the dataset,capturing thermal conditions that contribute to bleaching.

6.2 Observational Text Reports (AIMS)

The textual modality consists of coral reef monitoring reports obtained from the Australian Institute of Marine Science (AIMS). These documents contain descriptive, ground-level assessments of coral health such as:

1. bleaching symptoms (e.g “paling”, “patchy bleaching”)
2. environmental conditions (e.g “turbid water”, “low visibility”, “sedimentation”)
3. changes in coral cover or disease presence
4. reef location and date of observation

These unstructured textual descriptions provide qualitative insights that satellites cannot capture, making them an essential complementary data source.

6.3 Final Multimodal Dataset

For each day within the study period, satellite imagery and AIMS reports were matched based on date and region, producing a unified dataset. From each satellite image, numerical heat-stress features were extracted, while each observational report was converted into text embeddings. The combined dataset enables the model to learn from both physical ocean conditions and real-world ecological observations, supporting more reliable coral bleaching risk prediction.

CHAPTER 7

CONCLUSION OF CAPSTONE PROJECT PHASE - 1

7.1 Summary of Work Completed

Conducted literature review and overviewed it, which also included identification of data sources, and collected datasets. Analysed the problem statement and defined the scope of the project. Identified the research gap, and finalized the main objectives and outcomes from this capstone project.

7.2 Key Observations / Findings

Identified the insights gained from the literature review. Clearly identified and understood the main features affecting the reef health and causing coral bleaching. Potential predictive model identified. Identified the challenges or gaps in the current monitoring and prediction system.

CHAPTER 8

PLAN OF WORK FOR CAPSTONE PROJECT PHASE - 2

In the phase 2 of the capstone project, the focus will be mainly on building predictive models for coral bleaching. The work will begin with preprocessing of the dataset, by integrating datasets collected from NOAA (National Oceanic and Atmospheric Administration) and AIMS (Australian Institute of Marine Science), ensuring data consistency. Key environmental features which are mainly affecting the bleaching will be selected and engineered to improve model accuracy. Various predictive models will be, including machine learning models and deep learning models will be explored and evaluated using appropriate performance metrics. The final Model will be integrated with a visualization tool and early warning dashboards to highlight the areas of high risk reef bleaching. The project will be documented thoroughly, including methodology, results, and recommendations for sustainable reef management, culminating in a fully functional predictive system to support ongoing coral reef conservation efforts.

REFERENCES

[1] Yang, F., Liu, H., and Xu, Y. (2024). *Detection of Coral Reef Bleaching by Multitemporal Sentinel-2 Imagery*. Remote Sensing, 16(13), 2473. MDPI.

[2] Besson, S., Nebiker, S., and Kunz, M. (2024). *Coral Reef Monitoring with UAV-Based Object-Based Image Analysis*. ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, X-5-2024, 165–172.

[3] Alzahrani, S., Almutairi, A., Alotaibi, L., and Alshamrani, S. (2024). *Deep Learning-Based Coral Reef Condition Classification Using Enhanced Underwater Imagery*. Electronics, 13(24), 5027. MDPI.

[4] Shaha, S., Rathod, K., More, R., and Shelke, S. (2024). *Coral Reef Bleaching Prediction: A Machine Learning Approach Using Environmental Factors*. ResearchGate.

[5] González-Rivero, M., Hoegh-Guldberg, O., Bongaerts, P., Cortés, J., and Dove, S. (2024). *Environmental Drivers of Coral Bleaching under Thermal Stress*. Coral Reefs. Springer.

[6] Hensel, M. J. S., Oliver, T. A., Burden, C. M., Grossman, M., Felis, T., and Grottoli, A. G. (2025). *Long-Term Climate Variability and Its Role in Coral Reef Bleaching Patterns*. Earth System Dynamics, 16, 1503–1520.

[7] Ichikawa, S., Omori, A., Iwasaki, S., Yabe, T., and Sato, T. (2024). *Integrating Environmental Indicators and Multispectral Analysis for Coral Bleaching Assessment*. Coral Reefs. Springer.

APPENDIX A DEFINITIONS, ACRONYMS, ABBREVIATIONS

Coral Bleaching:
Process where corals turn white due to stress, primarily from warming ocean temperatures.

Zooxanthallae:
Microscopic algae living inside coral tissues that provide nutrients through photosynthesis and give corals their color.

Sea Surface Temperature (SST):
The temperature of the top layer of the ocean, widely used as an indicator of marine heat stress.

Degree Heating Weeks (DHW):
A cumulative measure of thermal stress experienced by corals over a 12-week period, used as a bleaching severity indicator.

Hotspot:
A satellite-derived metric indicating the intensity of ocean temperature anomalies above the long-term maximum monthly mean.

Remote Sensing:
The use of satellite-based instruments to observe, analyze, and collect data about Earth's surface and ocean conditions.

Predictive Modeling:

The development of statistical or machine-learning models to estimate future bleaching risks based on environmental inputs.

Acronyms & Abbreviations

Acronym	Full Form
SST	Sea Surface Temperature
DHW	Degree Heating Weeks
NOAA	National Oceanic and Atmospheric Administration
NLP	Natural Language Processing
ML	Machine Learning
AI	Artificial Intelligence
SST Anomaly	Sea Surface Temperature Anomaly
Chl-a	Chlorophyll-a Concentration
GIS	Geographic Information System
RGB	Red-Green-Blue (image channels)
API	Application Programming Interface